Question 10

10. Use the "USArrests" data and do as follows in the R studio with R markdown to knit PDF output:

a. Fit a hierarchical clustering model using single linkage and ge the dendogram for this model
b. Fit a hierarchical clustering model using complete linkage and get the dendogram for this model
c. Fit a hierarchical clustering model using average linkage and get the dendogram for this model
d. Show the number of clusters (k) to retain for the data using ablines in the dendogram of the best model

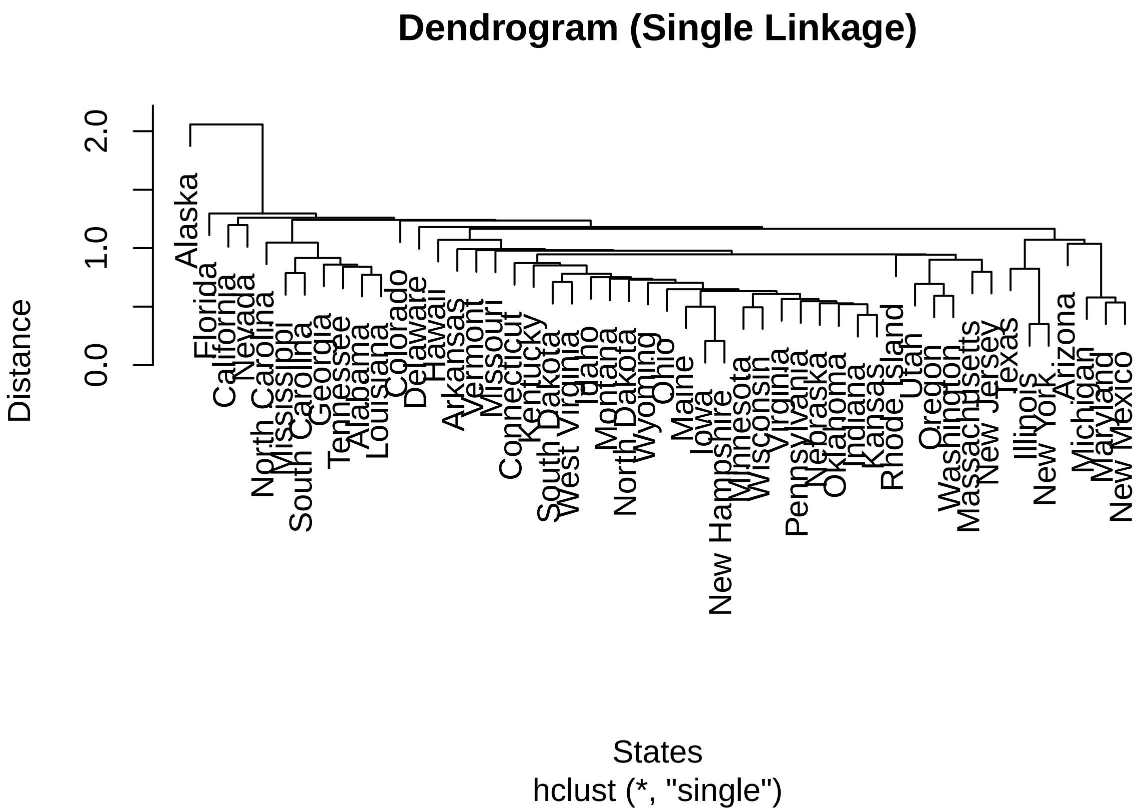
```
# Load the USArrests dataset
data(USArrests)

# Scale the data before clustering
df <- scale(USArrests)

# Calculate the distance matrix
dist_matrix <- dist(df, method = 'euclidean')
```

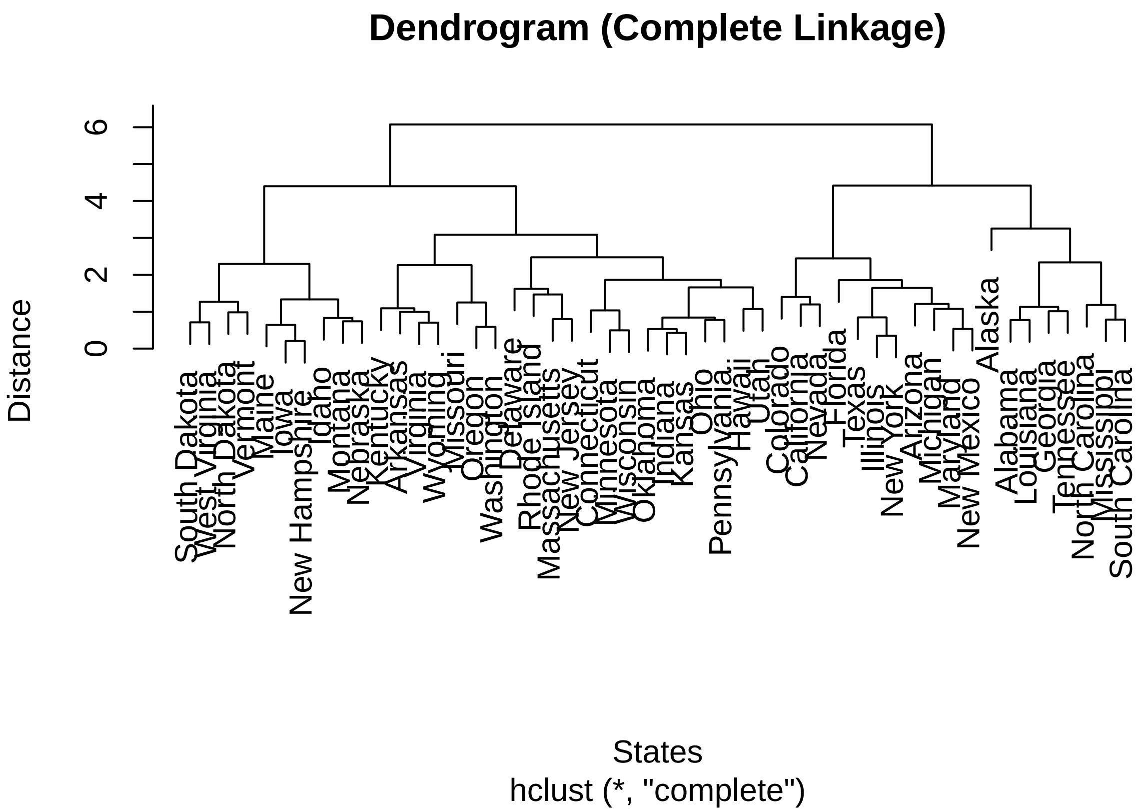
a. Hierarchical Clustering with Single Linkage

```
hc_single <- hclust(dist_matrix, method = "single")
plot(hc_single, main = "Dendrogram (Single Linkage)", xlab = "States", ylab = "Distance")
```



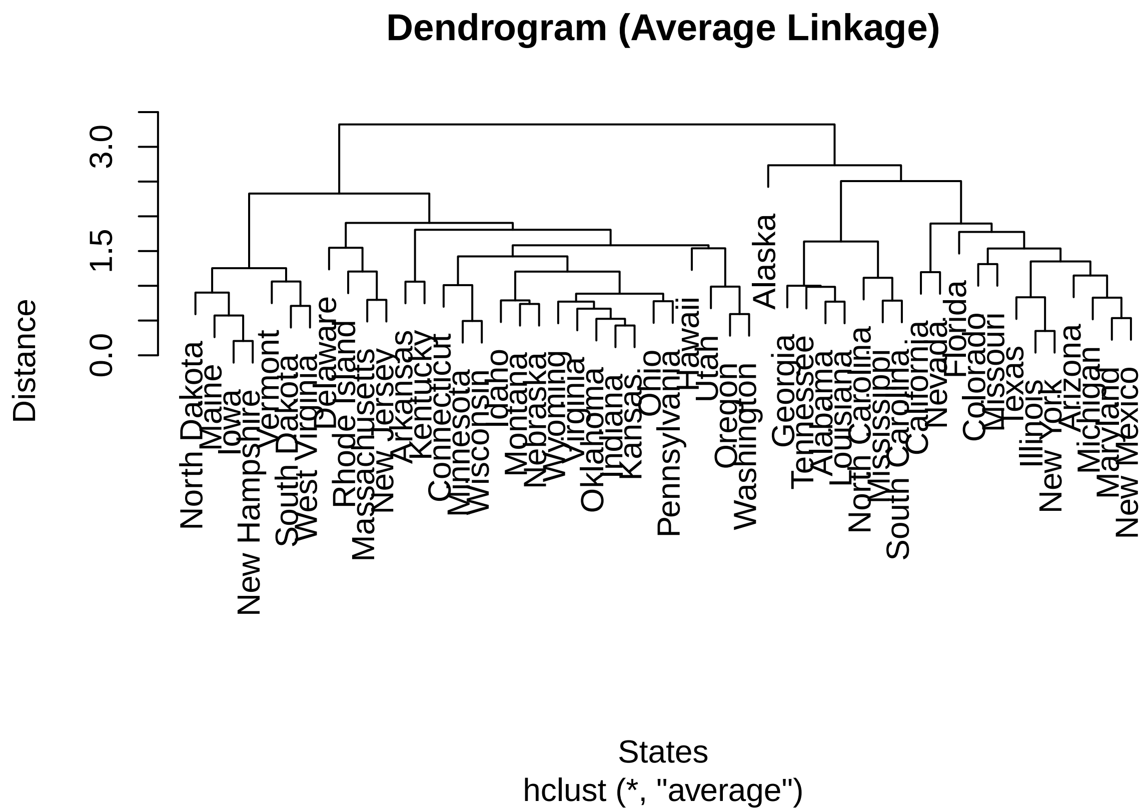
b. Hierarchical Clustering with Complete Linkage

```r
hc_complete <- hclust(dist_matrix, method = "complete")
plot(hc_complete, main = "Dendrogram (Complete Linkage)", xlab = "States", ylab = "Distance")
```



c. Hierarchical Clustering with Average Linkage

```r
hc_average <- hclust(dist_matrix, method = "average")
plot(hc_average, main = "Dendrogram (Average Linkage)", xlab = "States", ylab = "Distance")
```



d. Best Model and Number of Clusters

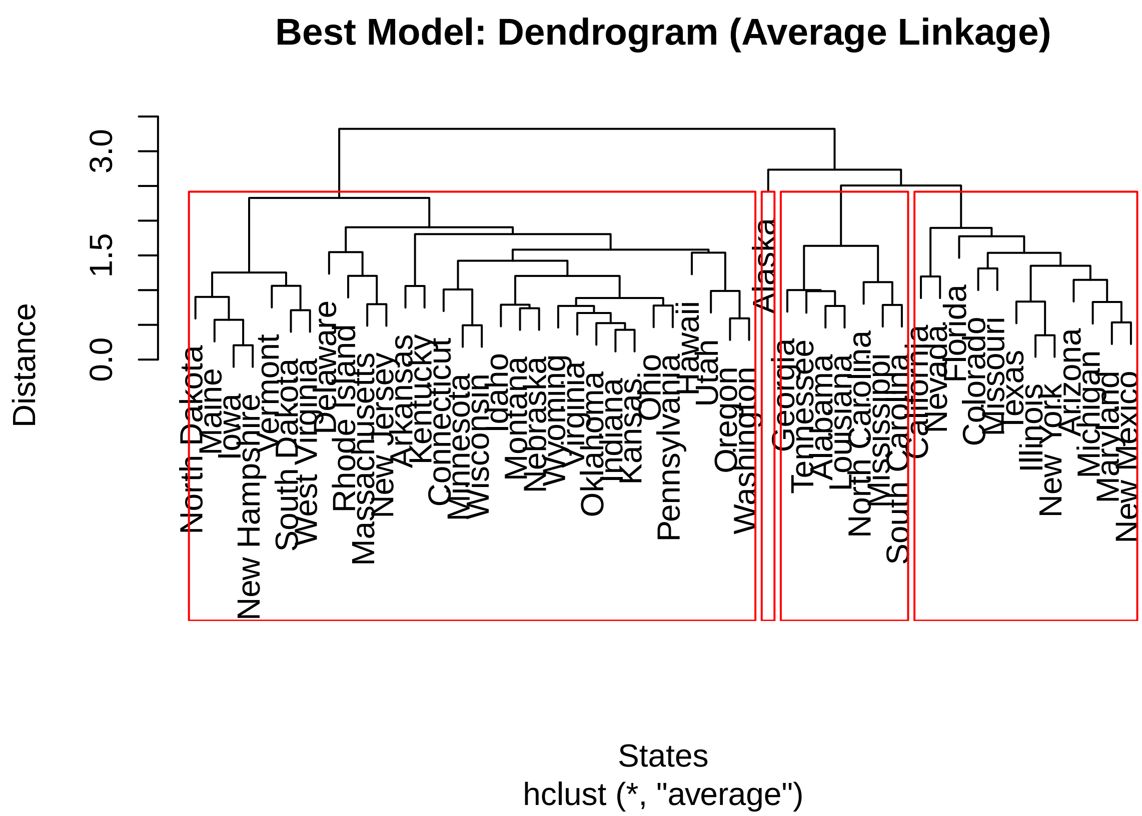
Choosing the best model: - **Single linkage** is prone to a "chaining" effect, where clusters are elongated and straggly. This is visible in its dendrogram. - **Complete linkage** tends to produce more compact, spherical clusters, but can be sensitive to outliers. - **Average linkage** is a compromise between the two. It is less susceptible to chaining and outliers.

For this analysis, we will choose the **average linkage** model as the "best" because it often provides a more balanced and interpretable clustering structure.

Determining the number of clusters (k): We can determine the number of clusters by examining the dendrogram and identifying the point where a horizontal cut would cross the most vertical distance without intersecting a cluster. Looking at the average linkage dendrogram, a cut at a height of around 3 would result in 4 distinct clusters. This is a common choice for this dataset.

```r
# Plot the dendrogram for the best model (average linkage)
plot(hc_average, main = "Best Model: Dendrogram (Average Linkage)", xlab = "States", ylab = "Distance")

# Add ablines to show k=4 clusters
rect.hclust(hc_average, k = 4, border = "red")
```



The red rectangles in the dendrogram above show the 4 clusters retained from the hierarchical clustering model with average linkage.